

PCSPF Pancreatic Cancer Survival — Complete Project Documentation

Technical Process Record, Methodology Audit, and Plain-Language Companion

Fundamentals of Data Science — CMSC 176

University of the Philippines Manila

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Primary artifact: PCSPF_Data_Science_Capstone.ipynb

Dataset: Dataset/PCSPF-Pancreatic Cancer Survival based on Preoperative Features.xlsx

Patients: 878 | Features: 20 | Random seed: 42

Table of Contents

1. Document Purpose and How to Read This Guide
2. Executive Overview
3. End-to-End Pipeline Map
4. Techniques and Methods Glossary
5. Repository and File Inventory
6. Complete Process Documentation (Sections 1–9)
7. Preprocessing Decisions Register
8. Visualizations Register (19 Figures)
9. Quantitative Results Reference
10. Compliance, Limitations, and Reproducibility

1. Document Purpose and How to Read This Guide

This document is the formal technical record of the PCSPF capstone project. Unlike a brief summary, it documents every major process step, the technique used, parameters applied, outputs produced, and decisions made—including steps deliberately not taken. Each notebook section (1 through 9) is expanded into process tables suitable for audit, defense preparation, and report writing.

How to read: Each process entry contains (a) a technical description for instructors and reviewers, and (b) an 'In simple terms' box for non-technical readers. Tables use wrapped text to prevent formatting overlap. All metrics match the executed notebook run unless noted as approximate.

2. Executive Overview

Problem: Predict 1-year pancreatic cancer survival and discover patient subgroups using 20 preoperative clinical features from 878 patients. Methods: Random Forest (supervised) and K-Means (unsupervised). Data were cleaned (3 columns removed), explored, preprocessed with documented decisions, split 80/20 stratified, clustered on scaled features (k=3), and classified with tuned Random Forest (class_weight=balanced).

In simple terms

We studied hospital lab and patient data to (1) estimate who is likely to live at least one year and (2) find natural groups of similar patients. We wrote down every step so others can verify and repeat the work.

Deliverable	Status	Location
Jupyter notebook (152 cells, 9 sections)	Complete & executed	PCSPF_Data_Science_Capstone.ipynb
Figures (19 PNG, 300 DPI)	Complete	figures/
Notebook builder script	Complete	scripts/assemble_notebook.py
This documentation PDF	v2.0	Documentations/PCSPF_Capstone_Project_Audit_Documentation.pdf
Separate written report	Not in repository	Student deliverable
Presentation slides	Not in repository	Student deliverable

3. End-to-End Pipeline Map

Phase	Notebook section	Key outputs
Phase A	Setup	Imports, seeds, figures/ folder
Phase B	Load & clean	Excel drop 3 cols survival_label 878×21
Phase C	EDA	Stats, 5 chart groups, correlation, class means
Phase D	Preprocess	Types, nulls, outliers, normality, VIF, decisions table
Phase E	Split & scale	702/176 stratified; scaler on full X for K-Means only
Phase F	Unsupervised	Elbow, silhouette, K-Means k=3, PCA, profiles, chi-square
Phase G	Supervised	Baseline RF, GridSearchCV, test metrics, importance
Phase H	Synthesis	Compare RF vs clusters, bridge table, limitations
Phase I	Conclusion	Dynamic metrics recap

In simple terms

The work flows left to right: prepare tools, clean data, explore, document choices, split data, find groups, build predictor, connect findings, summarize.

4. Techniques and Methods Glossary

Technique	What it is	How we used it	Key settings
Random Forest	Supervised ensemble of decision trees; each tree votes; handles non-linear patterns.	Predict 1-year survival from 20 labs.	class_weight=balanced; tuned via GridSearchCV.
K-Means	Unsupervised partitioning into k groups by minimizing within-cluster distance.	Discover patient subgroups.	Euclidean distance on StandardScaler features; k=3.
StandardScaler	Transforms each feature to mean 0, variance 1.	Fair distance for K-Means.	Fit on full X (878 rows) – unsupervised only.
PCA	Linear reduction to orthogonal components capturing variance.	Visualize 20-D clusters in 2D.	2 components; variance ratios reported.
GridSearchCV	Exhaustive search over hyperparameter grid with cross-validation.	Optimize RF settings.	108 combinations; 5 stratified folds; F1 score.
StratifiedKFold	CV folds preserve class proportions.	Fair tuning under imbalance.	5 splits, shuffle, random_state=42.
D'Agostino-Pearson	Normality test using skewness and kurtosis.	Test if features are bell-shaped.	normaltest(); alpha 0.05.
IQR outlier rule	Flag points outside $Q1-1.5*IQR$ or $Q3+1.5*IQR$.	Quantify extremes.	Counted but not removed.
VIF	Variance Inflation Factor; detects multicollinearity.	Check redundant columns.	All 20 features computed; none dropped.
Chi-square test	Tests independence of cluster vs survival labels.	Validate subgroup-outcome link.	chi2_contingency; p=0.2546.

5. Repository and File Inventory

File / path	Purpose
PCSPF_Data_Science_Capstone.ipynb	Main analysis: all code, markdown, outputs
Dataset/PCSPF-Pancreatic Cancer Survival based on Preoperative Features.xlsx	Source: 878 patients, 24 columns raw
figures/*.png	19 exported visualizations
scripts/assemble_notebook.py	Regenerates notebook programmatically
scripts/generate_audit_documentation.py	Regenerates this PDF
scripts/audit_content.py	Structured text for this PDF
requirements.txt	Python dependencies
Documentations/CMSC 176 Capstone Guidelines (Clean).pdf	Course blueprint

6. Complete Process Documentation (Sections 1–9)

The following subsections mirror the Jupyter notebook exactly. Every listed process step was implemented in code and executed. Process IDs (e.g., 4.5) match notebook subsection numbers.

SECTION 1: Project Setup and Imports

Established the computational environment, reproducibility controls, and plotting standards before any data were loaded.

In simple terms

We set up the software toolbox and rules so every chart and model result can be recreated exactly the same way later.

Process 1.1: Library imports

Process ID	1.1
Step name	Library imports
Technique / method	Python scientific stack
What we did (detailed)	Imported pandas and numpy for data; matplotlib and seaborn for visuals; scikit-learn for RandomForestClassifier, KMeans, PCA, StandardScaler, train_test_split, GridSearchCV, StratifiedKFold, classification metrics, silhouette_score; imbalanced-learn SMOTE (documented, not used); scipy chi2_contingency, shapiro, normaltest; statsmodels variance_inflation_factor.
Parameters / settings	N/A
Output / artifact	All dependencies loaded; warnings suppressed for clean output.
Decision / rationale	SMOTE imported only to document alternative imbalance strategy.

In simple terms

We loaded programs for tables, graphs, and machine learning.

Process 1.2: Reproducibility

Process ID	1.2
Step name	Reproducibility
Technique / method	Fixed random seed
What we did (detailed)	Set RANDOM_STATE = 42 on numpy, train_test_split, KMeans, RandomForest, StratifiedKFold, PCA. Set OPTIMAL_K = 3 as documented cluster choice.
Parameters / settings	random_state=42 everywhere
Output / artifact	Identical splits and model fits on re-run.
Decision / rationale	Mandatory for academic reproducibility.

In simple terms

We fixed the 'randomness dice' so results do not change every time.

Process 1.3: Visualization defaults

Process ID	1.3
Step name	Visualization defaults

Technique / method	matplotlib + seaborn
What we did (detailed)	Style: seaborn-v0_8-whitegrid; palette colorblind; figure 10x6; savefig dpi=300; font sizes 10-13pt.
Parameters / settings	dpi=300 for saved figures
Output / artifact	figures/ directory created with os.makedirs.
Decision / rationale	Publication-quality exports required by capstone spec.

In simple terms

We chose consistent colors and large, clear saved images for the report.

SECTION 2: Data Loading and Initial Inspection

Loaded raw Excel data, audited structure and quality, removed unusable columns, renamed target.

In simple terms

We opened the patient spreadsheet, checked it, and prepared a clean table for analysis.

Process 2.1: Load Excel

Process ID	2.1
Step name	Load Excel
Technique / method	pd.read_excel
What we did (detailed)	Read PCSPF workbook into df_raw. Verified shape (878, 24).
Parameters / settings	engine: openpyxl
Output / artifact	DataFrame 878 rows x 24 columns.
Decision / rationale	Single source file; no external augmentation.

In simple terms

We imported 878 patient rows from Excel.

Process 2.2: Structural inspection

Process ID	2.2
Step name	Structural inspection
Technique / method	Descriptive audit
What we did (detailed)	Displayed column names, dtypes, head(5), tail(5), isnull().sum(), duplicated().sum().
Parameters / settings	N/A
Output / artifact	Confirmed 878 nulls each in Predict label 1 and 0; 0 duplicates.
Decision / rationale	Predict columns unusable; ID all 'xxx'.

In simple terms

We listed columns, types, and missing values.

Process 2.3: Target distribution

Process ID	2.3
Step name	Target distribution
Technique / method	value_counts
What we did (detailed)	Class 1: 604 (68.8%); Class 0: 274 (31.2%); ratio 2.20:1.
Parameters / settings	N/A
Output / artifact	Imbalance documented for later class_weight.
Decision / rationale	Majority baseline accuracy = 68.8%.

In simple terms

About 7 in 10 patients survived at least one year in this dataset.

Process 2.4: Column removal

Process ID	2.4
Step name	Column removal
Technique / method	drop + rename
What we did (detailed)	Dropped ID, Predict label 1, Predict label 0. Renamed long label column to survival_label.
Parameters / settings	DROP_COLS list
Output / artifact	Clean shape (878, 21) = 20 features + target.
Decision / rationale	Asserted shape equals expected dimensions.

In simple terms

We removed useless columns and gave the outcome a short name.

SECTION 3: Exploratory Data Analysis (EDA)

Characterized distributions, class balance, correlations, and survival-group differences before modeling.

In simple terms

We explored the data with charts and tables to understand patterns before building models.

Process 3.1: Summary statistics

Process ID	3.1
Step name	Summary statistics
Technique / method	DataFrame.describe()
What we did (detailed)	Computed count, mean, std, min, quartiles, max for all 20 features.
Parameters / settings	Transposed for readability
Output / artifact	Table displayed in notebook.
Decision / rationale	Noted z-score scale (mean~0, std~1) in released file.

In simple terms

We calculated average, spread, and range of each measurement.

Process 3.2: Target bar chart

Process ID	3.2
Step name	Target bar chart
Technique / method	matplotlib bar + annotations
What we did (detailed)	Plotted class counts; labeled percents; saved target_distribution.png.
Parameters / settings	dpi=300
Output / artifact	Figure + printed imbalance ratio 2.20:1.
Decision / rationale	Foreshadowed class_weight strategy.

In simple terms

We charted how many survived vs did not.

Process 3.3: Feature histograms

Process ID	3.3
Step name	Feature histograms
Technique / method	seaborn histplot + KDE
What we did (detailed)	4x5 grid for 18 continuous features; saved feature_distributions.png.
Parameters / settings	30 bins
Output / artifact	Identified right-skewed markers (CEA, CRP, NLR, etc.).

Decision / rationale	Visual normality assessment before formal tests.
In simple terms	
We drew distribution shapes for each lab test.	

Process 3.4: Survival boxplots

Process ID	3.4
Step name	Survival boxplots
Technique / method	seaborn boxplot by class
What we did (detailed)	Grouped boxplots per feature by survival_label; saved boxplots_by_survival.png.
Parameters / settings	colorblind palette
Output / artifact	Modest separation on CA19-9, inflammation markers.
Decision / rationale	Weak overlap motivates multivariate models.

In simple terms

We compared lab values between survivors and non-survivors.

Process 3.5: Correlation heatmap

Process ID	3.5
Step name	Correlation heatmap
Technique / method	seaborn heatmap, mask upper triangle
What we did (detailed)	Full Pearson matrix; annotated r values; saved correlation_heatmap.png.
Parameters / settings	coolwarm/RdBu_r
Output / artifact	Max r with target < 0.15.
Decision / rationale	Strong inter-feature collinearity among ratios documented.

In simple terms

We mapped which variables move together.

Process 3.6: Multicollinearity narrative

Process ID	3.6
Step name	Multicollinearity narrative
Technique / method	Markdown analysis
What we did (detailed)	Documented NLR, PLR, SII, CRP/ALB formulas and bilirubin physiology.
Parameters / settings	N/A
Output / artifact	Implications stated for RF vs K-Means.
Decision / rationale	No code; conceptual section.

In simple terms

We explained why some columns repeat similar information.

Process 3.7: Class-conditional means

Process ID	3.7
Step name	Class-conditional means
Technique / method	groupby mean, diff table, bar chart
What we did (detailed)	Compared mean per feature class 0 vs 1; sorted by abs difference; saved class_conditional_means.png.
Parameters / settings	Top 10 features plotted
Output / artifact	Largest gaps: Abdominal Pain, CA19-9, Prealbumin.
Decision / rationale	Quantified weak univariate separation.

In simple terms

We measured which labs differ most between outcome groups.

Process 3.8: EDA summary

Process ID	3.8
Step name	EDA summary
Technique / method	Synthesis markdown
What we did (detailed)	Four themes: distributions, correlations, class differences, modeling implications.
Parameters / settings	N/A
Output / artifact	Informed preprocessing and model choice.
Decision / rationale	Single summary ties EDA to next sections.

In simple terms

We wrote what we learned before cleaning and modeling.

SECTION 4: Data Preprocessing

Documented every data decision including deliberate non-actions, with clinical and statistical justification.

In simple terms

We recorded every cleaning and preparation choice—and what we chose NOT to do and why.

Process 4.1: Data types

Process ID	4.1
Step name	Data types
Technique / method	dtype audit
What we did (detailed)	Verified Sex and Abdominal Pain binary int; 18 continuous float64; no casting needed.
Parameters / settings	unique() on binary cols
Output / artifact	All numeric and valid.
Decision / rationale	Stated in markdown confirmation cell.

In simple terms

We checked each column was the right number type.

Process 4.2: Missing values

Process ID	4.2
Step name	Missing values
Technique / method	isnull triple check
What we did (detailed)	Per-column sum, total sum, isna().any().any() — all zero on cleaned data.
Parameters / settings	N/A
Output / artifact	No imputation required.
Decision / rationale	Three independent verification methods.

In simple terms

We confirmed no empty cells in usable columns.

Process 4.3: Duplicates

Process ID	4.3
Step name	Duplicates
Technique / method	duplicated().sum()
What we did (detailed)	Found 0 exact duplicate rows; none removed.
Parameters / settings	N/A
Output / artifact	Each row treated as unique patient.

Decision / rationale	Documented in markdown.
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In simple terms

We checked no patient row was copied twice.

Process 4.4: Outliers

Process ID	4.4
Step name	Outliers
Technique / method	IQR rule 1.5x
What we did (detailed)	Per continuous feature: Q1, Q3, IQR, bounds, outlier counts and %; z-score boxplot figure.
Parameters / settings	1.5 * IQR
Output / artifact	outlier_boxplots.png; table displayed.
Decision / rationale	Decision: RETAIN all outliers (clinical + RF robust + sample size).

In simple terms

We counted extreme lab values but kept them because they matter medically.

Process 4.5: Skewness & normality

Process ID	4.5
Step name	Skewness & normality
Technique / method	skew, kurtosis, normaltest
What we did (detailed)	Skewness table with interpretation bands; D'Agostino-Pearson per feature with Normal? flag.
Parameters / settings	alpha=0.05
Output / artifact	Most features non-normal; shapiro imported not run.
Decision / rationale	Decision: NO log/Box-Cox/Yeo-Johnson.

In simple terms

We tested bell-curve assumptions and decided transforms were not needed.

Process 4.6: VIF

Process ID	4.6
Step name	VIF
Technique / method	variance_inflation_factor
What we did (detailed)	VIF for all 20 features; sorted table with low/moderate/high interpretation.
Parameters / settings	statsmodels VIF
Output / artifact	Elevated VIF on ratio features and components.
Decision / rationale	Decision: RETAIN all features.

In simple terms

We measured overlap between columns but kept all for clinical reasons.

Process 4.7: Feature engineering

Process ID	4.7
Step name	Feature engineering
Technique / method	None added
What we did (detailed)	Existing NLR, PLR, SII, CRP/ALB sufficient; rejected binning and interaction terms.
Parameters / settings	N/A
Output / artifact	Markdown justification only.
Decision / rationale	Avoid arbitrary information loss.

In simple terms

We did not create new calculated columns.

Process 4.8: Feature selection

Process ID	4.8
Step name	Feature selection
Technique / method	Retain all 20
What we did (detailed)	RF: use importance post-hoc; K-Means: all features with collinearity caveat.
Parameters / settings	N/A
Output / artifact	Consistent feature set across pipelines.
Decision / rationale	Same inputs for supervised and unsupervised.

In simple terms

We used all 20 measurements in both models.

Process 4.9: Class imbalance

Process ID	4.9
Step name	Class imbalance
Technique / method	class_weight strategy
What we did (detailed)	Documented weights ~1.60 class 0, ~0.73 class 1; rejected SMOTE.
Parameters / settings	balanced
Output / artifact	F1 and AUC as primary metrics.
Decision / rationale	Preserves real patients vs synthetic.

In simple terms

We told the model to pay more attention to the smaller group.

Process 4.10: Preprocessing summary

Process ID	4.10
Step name	Preprocessing summary
Technique / method	Table + figure
What we did (detailed)	11-row decision table rendered to preprocessing_summary.png.
Parameters / settings	matplotlib table
Output / artifact	Single reference for entire Section 4.
Decision / rationale	Consolidates audit trail.

In simple terms

We made one master table of all preparation decisions.

SECTION 5: Train-Test Split and Scaling

Separated X and y, stratified 80/20 split, applied scaling only for K-Means pipeline.

In simple terms

We divided data for fair testing and scaled numbers only for the grouping method.

Process 5.1: Feature-target split

Process ID	5.1
Step name	Feature-target split
Technique / method	X, y DataFrames
What we did (detailed)	X: 878x20 features; y: survival_label.
Parameters / settings	N/A
Output / artifact	Shapes printed.
Decision / rationale	Foundation for supervised pipeline.

In simple terms

We separated inputs from the outcome column.

Process 5.2: Train-test split

Process ID	5.2
Step name	Train-test split
Technique / method	train_test_split stratified
What we did (detailed)	test_size=0.2, stratify=y, random_state=42 train 702, test 176.
Parameters / settings	80/20
Output / artifact	Class proportions preserved in both sets.
Decision / rationale	20% holdout for unbiased RF evaluation.

In simple terms

We hid 20% of patients to test predictions later.

Process 5.3: StandardScaler

Process ID	5.3
Step name	StandardScaler
Technique / method	fit on full X for K-Means
What we did (detailed)	Before/after min, max, mean, std table; side-by-side boxplot figure.
Parameters / settings	StandardScaler()
Output / artifact	scaling_before_after.png.
Decision / rationale	RF explicitly NOT scaled.

In simple terms

We put all lab tests on the same scale for fair distance grouping.

SECTION 6: Unsupervised Learning — K-Means

Clustered patients on scaled features without survival_label; evaluated k; profiled clusters; bridged to outcomes.

In simple terms

We grouped similar patients by lab tests alone, then checked if those groups differed in survival.

Process 6.1: Elbow method

Process ID	6.1
Step name	Elbow method
Technique / method	KMeans inertia loop
What we did (detailed)	Fit k=2..10 on X_scaled; plot inertia vs k; elbow heuristic line.
Parameters / settings	n_init=10
Output / artifact	elbow_method.png.
Decision / rationale	Identifies diminishing returns in cluster count.

In simple terms

We found where adding groups stops helping much.

Process 6.2: Silhouette analysis

Process ID	6.2
Step name	Silhouette analysis
Technique / method	silhouette_score loop
What we did (detailed)	Mean silhouette per k; plot; max at k=2 (score ~0.206).
Parameters / settings	k=2..10
Output / artifact	silhouette_scores.png.
Decision / rationale	Compared with elbow for k choice.

In simple terms

We scored how tight and separated groups are.

Process 6.3: Choose k=3

Process ID	6.3
Step name	Choose k=3
Technique / method	Structured justification
What we did (detailed)	Silhouette favors k=2; chose k=3 for three clinical phenotypes and interpretability.
Parameters / settings	OPTIMAL_K=3
Output / artifact	Documented in markdown.

Decision / rationale	Trade-off statistical vs clinical granularity.
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In simple terms

We picked three patient types, not two.

Process 6.4: Final K-Means

Process ID	6.4
Step name	Final K-Means
Technique / method	fit_predict
What we did (detailed)	Labels for 878 patients; cluster size counts printed.
Parameters / settings	random_state=42, n_init=10
Output / artifact	df['cluster'] assigned.
Decision / rationale	Target never used in fit.

In simple terms

We assigned every patient to a group.

Process 6.5: PCA visualization

Process ID	6.5
Step name	PCA visualization
Technique / method	PCA n_components=2
What we did (detailed)	Explained variance printed; scatter by cluster; centroids marked.
Parameters / settings	PCA
Output / artifact	pca_clusters.png.
Decision / rationale	2D view of 20-D structure.

In simple terms

We drew a 2D picture of the groups.

Process 6.6: Cluster profiling

Process ID	6.6
Step name	Cluster profiling
Technique / method	groupby mean/std, heatmap
What we did (detailed)	Mean and std tables; z-scored heatmap of cluster profiles.
Parameters / settings	N/A
Output / artifact	cluster_profiles_heatmap.png.
Decision / rationale	Clinical interpretation of three phenotypes.

In simple terms

We described what each group looks like medically.

Process 6.7: Survival overlay

Process ID	6.7
Step name	Survival overlay
Technique / method	crosstab + chi-square
What we did (detailed)	Counts table; survival % per cluster; bar chart; chi2_contingency with expected freqs.
Parameters / settings	chi-square
Output / artifact	cluster_survival_overlay.png; p=0.2546.
Decision / rationale	Association not significant at 0.05.

In simple terms

We tested if groups differ in who survived.

Process 6.8: PCA by survival

Process ID	6.8
Step name	PCA by survival
Technique / method	Same PCA, color by label
What we did (detailed)	Compare cluster vs survival coloring in 2D.
Parameters / settings	N/A
Output / artifact	pca_survival_labels.png.
Decision / rationale	Bridge unsupervised to supervised outcome.

In simple terms

We overlaid true survival on the same 2D map.

SECTION 7: Supervised Learning — Random Forest

Binary classification of 1-year survival with balanced Random Forest, grid search, and full evaluation.

In simple terms

We built a prediction tool for survival and measured how well it works.

Process 7.1: Baseline RF

Process ID	7.1
Step name	Baseline RF
Technique / method	RandomForestClassifier
What we did (detailed)	100 trees, class_weight balanced; fit X_train; evaluate X_test.
Parameters / settings	n_estimators=100
Output / artifact	Baseline acc 68.18%; CM and report printed.
Decision / rationale	Benchmark before tuning.

In simple terms

We ran a first draft model for comparison.

Process 7.2: GridSearchCV

Process ID	7.2
Step name	GridSearchCV
Technique / method	Hyperparameter search
What we did (detailed)	Grid: n_estimators [100,200,300], max_depth [None,10,15,20], min_samples_split [2,5,10], min_samples_leaf [1,2,4]; 108 combos; 5-fold stratified CV; scoring=f1.
Parameters / settings	StratifiedKFold(5)
Output / artifact	Best: 300 trees, depth 15; CV F1 ~0.809.
Decision / rationale	Top 10 combos table displayed.

In simple terms

We automatically tried many settings and picked the best.

Process 7.3: Test evaluation

Process ID	7.3
Step name	Test evaluation
Technique / method	Metrics suite
What we did (detailed)	Accuracy vs baselines; classification report figure; confusion matrix; ROC AUC 0.613; metrics table.
Parameters / settings	holdout 176 patients
Output / artifact	4 figures saved.

Decision / rationale	Discussed FN vs FP clinically.
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In simple terms

We scored final performance on unseen patients.

Process 7.4: Feature importance

Process ID	7.4
Step name	Feature importance
Technique / method	feature_importances_
What we did (detailed)	Horizontal bar chart all 20 features; top 5 and bottom 5 printed.
Parameters / settings	Gini importance
Output / artifact	feature_importance.png.
Decision / rationale	CEA, Prealbumin, CA19-9 lead.

In simple terms

We ranked which labs mattered most for predictions.

SECTION 8: Synthesis and Bridging Analysis

Connected supervised importances, unsupervised structure, and predicted probabilities.

In simple terms

We compared prediction results with patient groups to see if both analyses tell the same story.

Process 8.1: Importance vs differentiation

Process ID	8.1
Step name	Importance vs differentiation
Technique / method	Rank comparison
What we did (detailed)	RF importance rank vs cluster mean-range rank; side-by-side chart.
Parameters / settings	N/A
Output / artifact	synthesis_comparison.png.
Decision / rationale	Overlap in top features = convergence.

In simple terms

We checked if important predictors also define groups.

Process 8.2: Probability by cluster

Process ID	8.2
Step name	Probability by cluster
Technique / method	predict_proba on full X
What we did (detailed)	Mean predicted survival probability vs actual rate per cluster.
Parameters / settings	tuned RF
Output / artifact	Bridge table with difference column.
Decision / rationale	Calibration across subgroups.

In simple terms

We compared model confidence in each group.

Process 8.3: Clinical synthesis

Process ID	8.3
Step name	Clinical synthesis
Technique / method	Dynamic narrative
What we did (detailed)	High-risk cluster 2 vs low-risk cluster 0; convergence paragraph.
Parameters / settings	runtime vars
Output / artifact	Rendered from executed results.
Decision / rationale	Mutual validation narrative.

In simple terms

We wrote the medical story connecting both analyses.

Process 8.4: Limitations

Process ID	8.4
Step name	Limitations
Technique / method	10 items
What we did (detailed)	Imbalance, weak correlations, no external validation, K-Means assumptions, etc.
Parameters / settings	N/A
Output / artifact	Numbered list.
Decision / rationale	Honest scope boundaries.

In simple terms

We listed what the study cannot claim.

Process 8.5: Future work

Process ID	8.5
Step name	Future work
Technique / method	10 items
What we did (detailed)	XGBoost, DBSCAN, transforms, SHAP, Cox models, external validation, staging features, etc.
Parameters / settings	N/A
Output / artifact	Research directions listed in notebook markdown.
Decision / rationale	Not implemented in current scope.

In simple terms

We listed next steps for stronger research if the project continues.

SECTION 9: Conclusion

Four-paragraph recap with metrics injected from runtime variables.

In simple terms

The ending pulls real numbers from the run so the summary always matches results.

Process 9.1: Dynamic conclusion

Process ID	9.1
Step name	Dynamic conclusion
Technique / method	IPython.display.Markdown f-string
What we did (detailed)	Paragraphs: objective, supervised results, unsupervised results, bridging insight.
Parameters / settings	tuned_acc, roc_auc, macro_f1, top_feats, surv_rate, p
Output / artifact	KEY RESULTS print block + rendered markdown.
Decision / rationale	Avoids stale hard-coded metrics.

In simple terms

The final summary auto-fills accuracy, AUC, clusters, and p-value.

7. Preprocessing Decisions Register

Complete list of preprocessing actions from Section 4.10. Each row is a deliberate project decision.

Step	Action	Justification
Drop ID, Predict labels	Removed	ID anonymized; predict columns 100% NaN
Missing values	No imputation	Zero nulls after column drop
Duplicates	None removed	0 duplicate rows
Outliers	Retained	Clinically meaningful extremes
Normality transform	Not applied	RF rank-invariant; scaler for K-Means
Multicollinearity	Retain all features	RF subsampling; clinical indices
Feature engineering	None added	Existing ratios sufficient
SMOTE	Not used	class_weight preferred
RF scaling	None	Trees scale-invariant
K-Means scaling	StandardScaler	Distance-sensitive
Train-test split	80/20 stratified	702 train / 176 test

8. Visualizations Register (19 Figures)

#	Filename	Section	Description
1	target_distribution.png	3.2	Bar chart of class 0 vs class 1 with count and percent labels
2	feature_distributions.png	3.3	4x5 grid of histograms + KDE for 18 continuous features
3	boxplots_by_survival.png	3.4	Boxplots of each continuous feature by survival label
4	correlation_heatmap.png	3.5	Lower-triangle Pearson correlation heatmap (20 features + target)
5	class_conditional_means.png	3.7	Horizontal bar chart of top 10 mean differences by class
6	outlier_boxplots.png	4.4	Per-feature z-score boxplots for outlier comparison
7	preprocessing_summary.png	4.10	Table figure of all preprocessing decisions
8	scaling_before_after.png	5.3	Side-by-side boxplots before/after StandardScaler
9	elbow_method.png	6.1	Line plot of inertia vs k (2-10) with elbow marker
10	silhouette_scores.png	6.2	Line plot of mean silhouette vs k
11	pca_clusters.png	6.5	2D PCA scatter colored by cluster + centroids
12	cluster_profiles_heatmap.png	6.6	Heatmap of z-scored mean features per cluster
13	cluster_survival_overlay.png	6.7	Grouped bar chart of survival counts per cluster
14	pca_survival_labels.png	6.8	2D PCA scatter colored by true survival
15	confusion_matrix.png	7.3	Seaborn heatmap of tuned RF predictions
16	roc_curve.png	7.3	ROC with AUC annotation and diagonal reference
17	feature_importance.png	7.4	Horizontal bar chart of all 20 Gini importances
18	classification_report.png	7.3	Tabular precision/recall/F1 figure
19	synthesis_comparison.png	8.1	Side-by-side top-10 RF vs cluster-differentiation

9. Quantitative Results Reference

9.1 Dataset and split

Metric	Value
Patients	878
Features	20
Target	survival_label (1 = survived >= 1 year)
Train / test	702 / 176 (80/20 stratified)
Class 1 / Class 0	604 (68.8%) / 274 (31.2%)
Imbalance ratio	2.20 : 1

9.2 Supervised learning (Random Forest)

Metric or item	Value	Notes
Majority-class baseline	68.8%	Always predict survived
Untuned RF accuracy	68.18%	100 trees, before grid search
Tuned RF test accuracy	71.02%	+2.27 pp vs baseline
Macro F1 (test)	0.507	Both classes combined
AUC-ROC (test)	0.613	Ranking ability
Best CV F1 (mean)	~0.809	On training folds only
Best n_estimators	300	GridSearch winner
Best max_depth	15	GridSearch winner
Best min_samples_split	2	GridSearch winner
Best min_samples_leaf	1	GridSearch winner
Top 5 features (importance)	CEA, Prealbumin, CA19-9, Total Bilirubin, BMI	Gini impurity decrease

In simple terms

The prediction model beats always guessing 'survived' by about 2 percentage points on accuracy. It is only modestly better at ranking risk (AUC near 0.61). CEA, prealbumin, and CA19-9 are among the strongest inputs.

9.3 Unsupervised learning (K-Means, k=3)

Cluster	N	% cohort	Survival rate	Clinical profile (summary)
0	519	59%	70.1%	Lower inflammation; higher prealbumin/ALB
1	204	23%	69.6%	Elevated bilirubin, NLR, SII; lower prealbumin
2	155	18%	63.2%	Highest CRP/CRP/ALB; more abdominal pain

Chi-square test (cluster vs survival): $p = 0.2546$ (not significant at $\alpha = 0.05$). Cluster 2 has the lowest survival (63.2%) and highest inflammatory profile.

10. Compliance, Limitations, and Reproducibility

10.1 Critical rules compliance

Rule	Implementation evidence	Status
No supervised leakage via scaler	StandardScaler fit on full X for K-Means only; RF uses unscaled train/test	Met
K-Means excludes target	fit_predict on X_scaled only; survival overlaid post-hoc	Met
Drop unusable columns	ID, Predict label 1, Predict label 0 removed	Met
RF not scaled	X_train, X_test raw	Met
K-Means scaled	StandardScaler on X	Met
Beyond accuracy	F1, AUC, CM, classification report	Met
class_weight balanced	All RandomForestClassifier instances	Met
random_state = 42	Split, models, CV, PCA	Met
Normality testing	D'Agostino-Pearson on all 18 continuous features	Met
Shapiro-Wilk	Imported; not executed (see Process 4.5)	Partial

10.2 Known limitations

1. Moderate class imbalance (2.2:1) despite balanced weights.
2. Weak univariate correlations (all $|r| < 0.15$) — interactions required.
3. Single internal test split; no external hospital validation.
4. K-Means assumes spherical, similar-variance clusters.
5. Multicollinearity among NLR, PLR, SII, CRP/ALB and components.
6. AUC 0.613 — limited discrimination for clinical screening.
7. Features in released file are pre-standardized (z-scores).
8. Retrospective data — association, not causation.
9. Missing tumor staging and treatment variables.

10.3 Reproducibility

1. pip install -r requirements.txt
2. Open terminal in project root (contains Dataset/ and figures/)
3. jupyter notebook open PCSPF_Data_Science_Capstone.ipynb
4. Kernel Restart & Run All (~2–5 min for grid search)
5. Regenerate notebook: python scripts/assemble_notebook.py
6. Regenerate this PDF: python scripts/generate_audit_documentation.py

End of document. This record reflects the executed capstone pipeline and is intended for audit, reporting, and oral defense preparation.